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1. Introduction to ALMA

Science & Capabilities

The Atacama Large Millimeter/submillimeter Array (ALMA) is currently the world largest radio interferometer, consisting of 66 antennas with 12 m and 7 m diameter, located at 5050 m on the Chajnantor Plateau in the Atacama Desert of Chile. With antenna separations (or baselines) of up to 16 km, ALMA can reach exceptional angular resolutions of a few milliarcseconds (0.005″ at 950 GHz). ALMA operates across ten high-frequency receiver bands, spanning 35 to 950 GHz and covering the mm and submm regimes (see Figure 1).

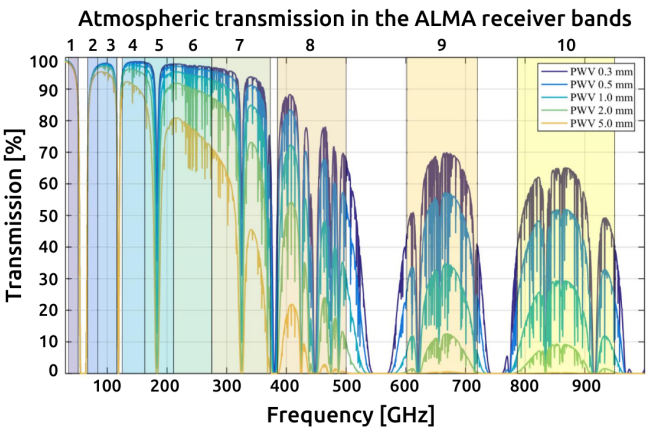

Figure 1: Atmospheric transmission across the ten ALMA receiver bands (from Carpenter et al. 2020). The curves correspond to different amounts of precipitable water vapor (PWV), which is the total amount of water vapor integrated along the atmospheric column. ALMA bands are placed in the atmospheric windows where transmission is highest. At higher frequencies, especially Bands 8 to 10, very dry conditions become essential for the observations.

ALMA is an extremely versatile facility, providing high-fidelity imaging, sensitive spectroscopy to study molecular gas and chemistry, and continuum observations to trace dust and star formation. It also offers polarization capabilities to probe magnetic fields, and can be used in VLBI mode to reach event-horizon-scale resolution, as demonstrated by the Event Horizon Telescope (EHT Collaboration et al. 2019).

With the upcoming Wideband Sensitivity Upgrade (WSU, Carpenter et al. 2022), ALMA will enter a new phase of enhanced sensitivity, increased observing speed, and greater calibration flexibility, making this an ideal time to learn how to design observing programs and process ALMA data.

ALMA Observations & Data

Because sky frequencies are too high to be processed directly by the receiver electronics, the sky signal is first converted to a lower frequency using a mixer and a very stable reference frequency called Local Oscillator (LO). As a result, most of the ALMA bands are heterodyne receivers with two sidebands (see Figure 2). Band 1 is the only single-sideband receiver.

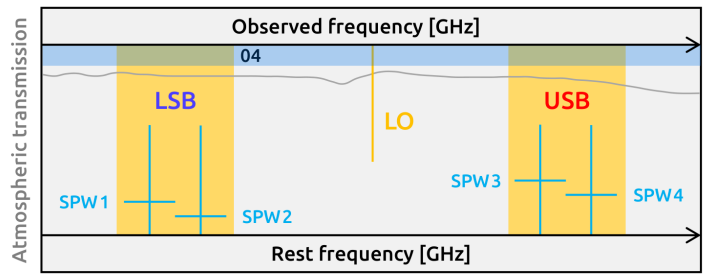

Figure 2: Illustration of a spectral setup using the ALMA Band 4 receiver in the Observing Tool (OT). The observed frequency axis is split into a lower sideband (LSB) and an upper sideband (USB) around the local oscillator (LO), a stable reference frequency. Four spectral windows (SPWs) are distributed across the two sidebands. The solid curve represents the atmospheric transmission.

ALMA observations can use up to four spectral windows (SPWs), distributed between the lower sideband (LSB) and upper sideband (USB) around the LO frequency (Figures 2 and 3), enabling simultaneous continuum and line imaging. Each SPW is processed by the ALMA correlator and can be configured in Time Division Mode or Frequency Division Mode, the latter providing high spectral resolution.

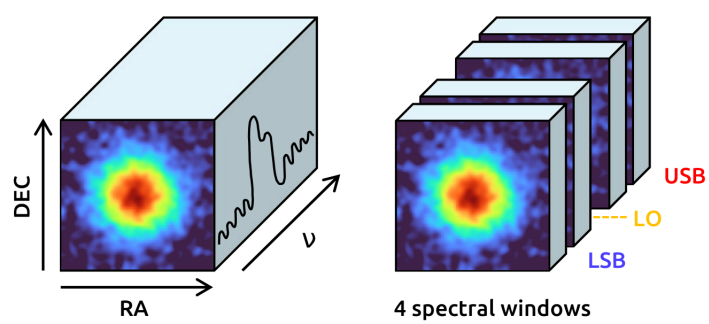

Figure 3: *Left:* Representation of a mock ALMA datacube with two spatial angular dimensions (RA, Dec) and one spectral dimension (frequency axis ν), with the emission spectrum illustrated. *Right:* Four datacubes corresponding to four spectral windows, distributed between the lower sideband (LSB) and upper sideband (USB) around the local oscillator (LO) frequency (see Figure 2).

⇒ The following sections present concise notes introducing the main steps of ALMA data reduction using CASA.

2. Running the Pipeline

In this tutorial, we use the ALMA program 2015.1.01559.S. After downloading the data from the ALMA Archive and the correct CASA version for running the cycle-related pipeline, go to the directory where the script for the PI is stored, which is needed to generate the calibrated *uv* products:

```
>>> cd 2015.1.01559.S/science_goal[.]/group[.]/
    member[.]/script
# [.] ≡ uid code
```

Then launch CASA specifying the pipeline is run:

```
>>> casa453 --pipeline
# casa453 ≡ alias for launching CASA 4.5.3
>>> SPACESAVING=2
# Parameter that controls the use of disk space
during data processing.
>>> execfile("ScriptForPI.py")
# Script to generate calibrated uv products.
```

Once the process is complete, the calibrated *uv* products are stored in the output directory `calibrated` with an extension `.ms.split.cal`. For imaging, it is not necessary to use the pipeline-specific CASA version (e.g., use the latest version).

3. Image Properties

The LISTOBS task provides all the essential data information for imaging, which is stored in the measurement sets (`.ms`).

```
>>> listobs(vis=[filename].ms.split.cal)
# vis ≡ visibilities are uv-plane measurements
```

To begin, we will determine the imaging parameters `imsize` and `cell`, which are specific to each ALMA observation.

The primary beam (FOV) is computed as

$$\theta_{\rm PB}['] = \theta_{\rm ref} \times \frac{\nu_{\rm ref}}{\nu_{\rm obs}} \tag{1}$$

For the 12-m array[1], $\theta_{\rm ref} = 19''$ at $\nu_{\rm ref} = 300$ GHz (1 mm). In our example, $\theta_{\rm PB} = 62''$ in Band 3 at $\nu_{\rm obs} = 91.5$ GHz.

The synthesized beam (PSF) is computed as

$$\theta_{\rm SB}['] = \frac{76}{{\rm B_{max}[km]} \times \nu_{\rm obs}[{\rm GHz}]} \tag{2}$$

We compute $\theta_{\rm SB} = 0''.55$ at 91.5 GHz and for $B_{\rm max} = 1.5$ km. To sample this scale with at least five pixels, we set a cell size of $0''.1$. The image size is given by twice $\theta_{\rm PB}$ (i.e., twice the FWHM) divided by the cell size, yielding a 1240-pixel image. To speed up imaging, it is common to crop the field. Here, we use a 300-pixel box, sufficient to cover the source.

[1] ALMA Basics

4. Multi-Frequency Synthesis Imaging

MFS imaging consists of gridding visibilities from multiple frequency channels onto a single spatial-frequency grid using the TCLEAN task. The resulting map contains both line and continuum emission. In this science case, two visibility files (corresponding to two observing runs) are combined.

```
>>> tclean(vis=["[...].ms.split.cal",
"[...].ms.split.cal"], field=["2","2"],
imagename="B3_MFS", imsize=[300],
cell=["0.1arcsec"], phasecenter="J2000
06h58m37.559 -55d57m05.075", specmode="mfs")
```

Then, we use `specmode="cube"` to generate a datacube containing the four SPWs and to identify the spectral channels where line emission is present. Blank channels correspond to frequency gaps between the SPWs (see Figure 3).

5. Continuum Imaging

We must select spectral channels free of line emission. It is recommended to first run TCLEAN in interactive mode using natural weighting. Then, run it in non-interactive mode using the output `.mask` image and increase the angular resolution by applying Briggs weighting with a robustness parameter.

```
>>> tclean(vis=["[...].ms.split.cal",
"[...].ms.split.cal"], field=["2","2"],
spw=["0:0~96;126~239,1:0~159;191~239,2,3"],
imagename="B3_CONT_CLEAN_3xRMS",
imsize=[300], cell=["0.1arcsec"],
phasecenter="J2000 06h58m37.559 -55d57m05.075",
specmode="mfs", nchan=-1, restfreq=["91.5GHz"],
pbcor=True, weighting="natural",
niter=10000, threshold="24e-6Jy",
interactive=True, mask="")
```

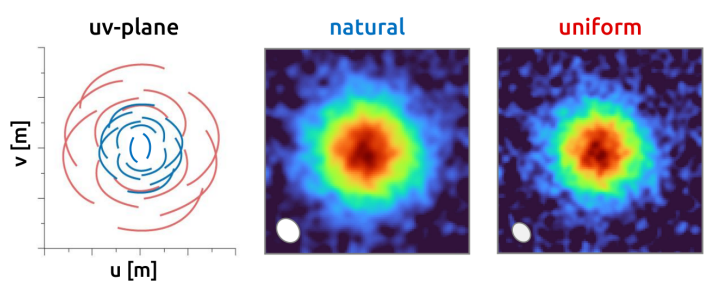

Figure 4: Illustration of *uv*-plane sampling and two data-weighting schemes: the natural weighting emphasizes dense *uv*-plane regions (short baselines), while uniform weighting emphasizes extended baselines (outer *uv*-plane). The ellipses are the corresponding synthesized beams. Natural weighting enhances the signal-to-noise ratio at the expense of angular resolution, with the opposite effect for uniform. In practice, we often use Briggs weighting, which uses a robustness parameter (R) to provide a compromise between the two.

6. Continuum Subtraction

Here, the spectral line is CO(3–2). We subtract the continuum from the *uv*-plane using the task UVCONTSUB, for which we indicate the spectral windows containing the line (see `spw`), and ask to fit the spectrum where only continuum emission is present by specifying the channel ranges (see `fitspec`). The task UVCONTSUB generates a new visibility dataset in the `dir_cal` directory, having the extension `.uvcontsub`. This process is done separately for each measurement set.

```
>>> uvcontsub(vis="[...].ms.split.cal",
field="2", spw="0,1", datacolumn="data",
fitspec="0:0~96;126~239,1:0~159;191~239,2,3")
```

7. Spectral Line Imaging

We must select spectral channels covering the line emission. It is recommended to generate a datacube with a total velocity width of 2000 km s^{-1} that is centered on the spectral line. We set `restoringbeam` to `common` to ensure that all channels are convolved to the same synthesized beam, enabling consistent spatial resolution across the line cube. As for continuum imaging, we first run TCLEAN in interactive mode applying natural weighting, and then in non-interactive mode using the output `.mask` image and Briggs weighting.

```
>>> tclean(vis=["[...].ms.split.cal.uvcontsub",
"[...].ms.split.cal.uvcontsub"],
field=["2", "2"], spw=["0,1"],
imagename="B3_CUBE_CO32_CLEAN_3xRMS",
imsize=[300], cell=["0.1arcsec"],
specmode="cube", nchan=80, start="-1000km/s",
width="26km/s", restfreq=["91.5GHz"],
deconvolver="hogbom", restoringbeam="common",
pbcor=True, weighting="natural",
niter=10000, threshold="24e-6Jy",
interactive=True, mask="")
```

8. Line Moment Maps

Moments are 2D maps derived from a 3D line cube by taking statistical moments along the frequency (velocity) axis. The task used to generate moment maps is IMMOMENTS.

Moment 0 is the velocity-integrated flux [Jy beam^{-1} km s^{-1}], providing the spatial distribution of line emission within the source. We select only the channels containing line emission (see `chans`), and include all the pixels (`includepix=-1`).

```
>>> immoments(imagename="[...].image",
moments=[0], axis="spectral", chans="34~55",
stokes="I", includepix=-1, excludepix=-1,
outfile="[...].image.mom0")
```

Moment 1 maps the rotation velocity [km s^{-1}], and moment 2 the velocity dispersion [km s^{-1}]. The RMS noise per channel must be taken into account to reject noisy data. Here, pixels below $4\sigma \simeq 0.7$ mJy are excluded (see `includepix`).

```
>>> immoments(imagename="[...].image",
moments=[1], axis="spectral", chans="34~55",
stokes="I", includepix=[0.0007, 1e10],
excludepix=-1, outfile="[...].image.mom1")
```

Figure 5 shows simulations of the first three moment maps of the mock ALMA line cube shown in Figure 3. The moment maps share the same averaged synthesized beam as the parent line cube from which they are extracted. The blank regions in moments 1 and 2 correspond to masked (noisy) data.

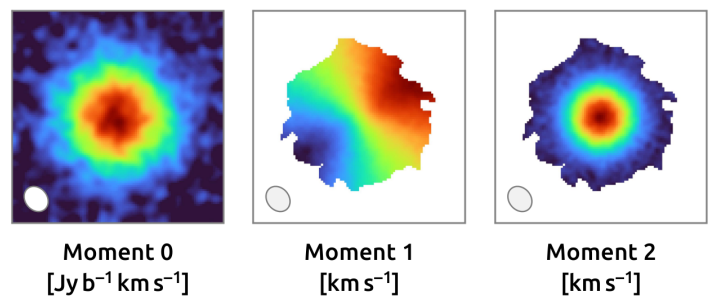

Figure 5: Illustration of the first moment maps of a mock line cube (Figure 3). Moment 0 maps the velocity-integrated flux, moment 1 the rotation velocity field, and moment 2 the velocity dispersion.

Converting to FITS

EXPORTFITS converts the CASA products into FITS files.

```
>>> exportfits(imagename="[...].image.pbcor",
fitsimage="[...].fits", dropdeg=False)
# dropdeg=True drops all degenerate axes, i.e.,
the 3rd axis (frequency) and 4th axis (stokes).
```

APPENDIX

- Illustrated guidelines for preparing an ALMA proposal: ALMA Science Primer (Cycle 12)
- Detailed ALMA tutorials for data reduction
- List of all CASA tasks for data processing
- Data visualization and analysis with CARTA
- Schematic figures (Credit: C. Cornil-Baïotto)

C. Cornil-Baïotto
carla.cornil@postgrado.uv.cl